Reference Coordinate Systems (3/3)

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Reference Coordinate Systems

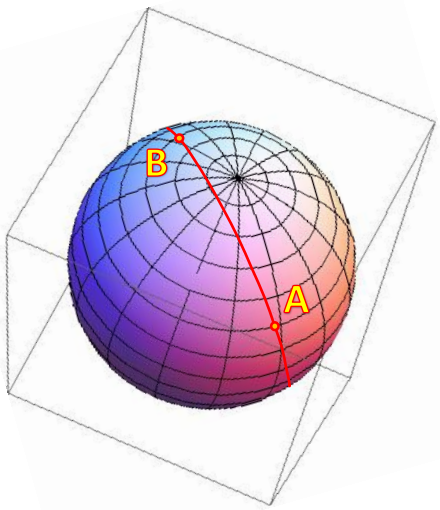
- ECEF Coordinate System
- WGS-84 and PZ-90 Frames
- Geodetic Coordinates
- Geodetic to Cartesian Coordinates
- Cartesian to Geodetic Coordinates
- The Geoid
- Reference Ellipsoids
- Local Geodetic Datums
- Datum Transformations
- Molodensky Transformation
- Rotation Matrices
- ENU Coordinates
- Azimuth & Elevation
- Orthodromes
- Loxodromes

ECEF – Earth-Centered, Earth-Fixed
WGS – World Geodetic System
PZ – Parametry Zemli (Russian)
ENU – East-North-Up

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

- Orthodromes are the shortest path, on the surface of sphere, between two points;
- Orthodromes follow great circles;
- Great circles share the centre with the sphere;
- Heading changes along orthodromes;
- Orthodromes are straight lines in a gnomonic projection.

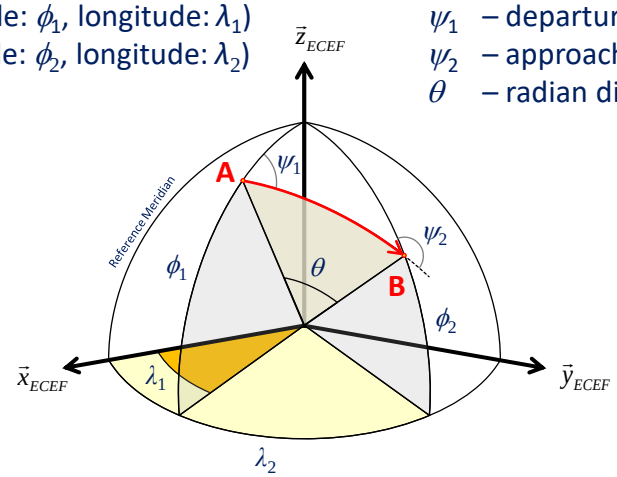


Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
ψ_2 – approach heading
θ – radian distance



$r \cdot \theta$ – orthodrome length
r – Earth radius

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
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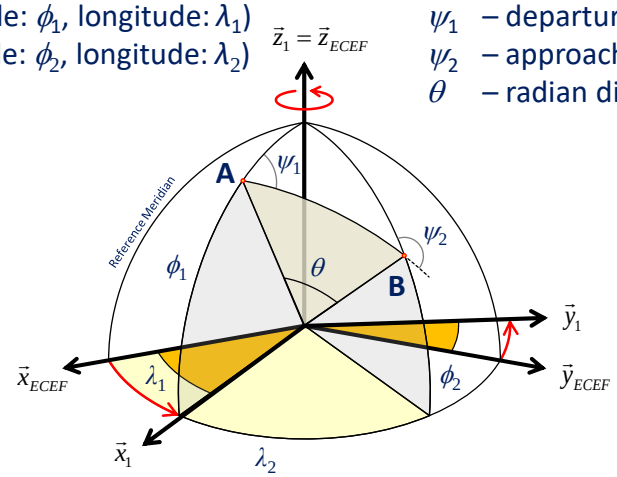

ECEF→Rot$_z$(λ_1)→Ref.1

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
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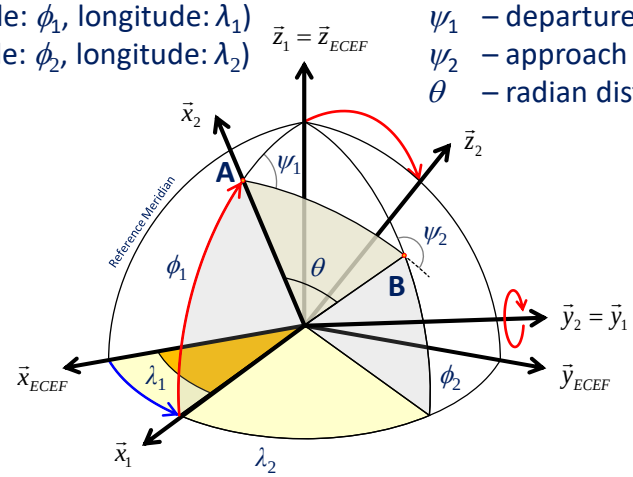
θ – radian distance



ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Ref.2

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
ψ_2 – approach heading
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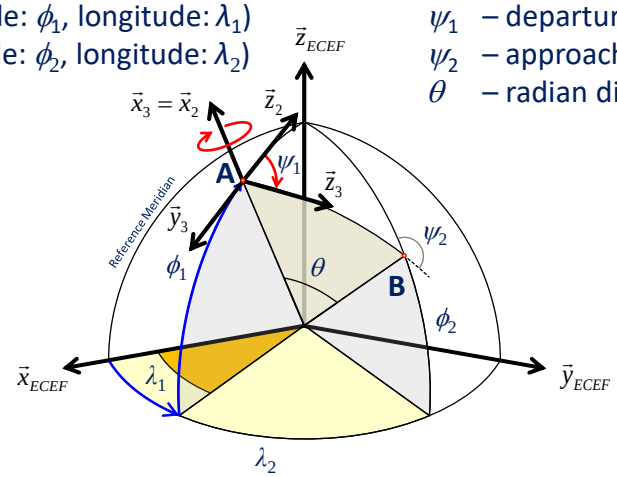

ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Ref.3

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
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ψ_1 – departure heading
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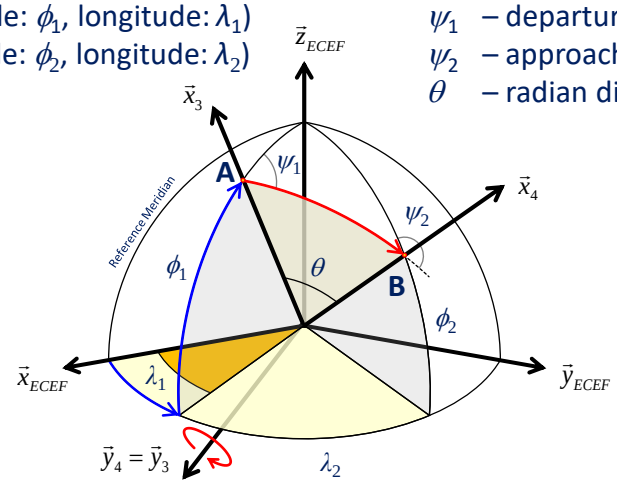


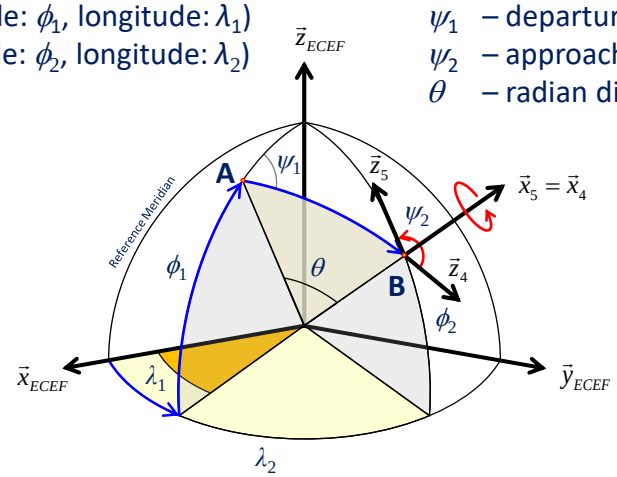
ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Rot$_y$(-θ)→Ref.4

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
ψ_2 – approach heading
θ – radian distance



ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Rot$_y$(-θ)→Rot$_x$(ψ_2)→ Ref.5

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

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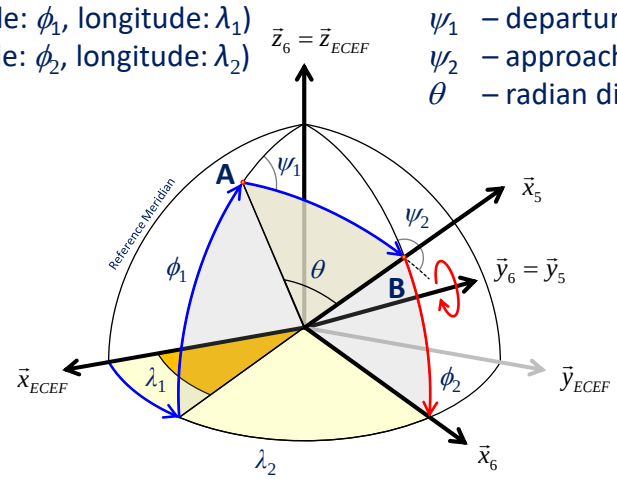

ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Rot$_y$(-θ)→Rot$_x$(ψ_2)→ Rot$_y$(ϕ_2)→Ref.6

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

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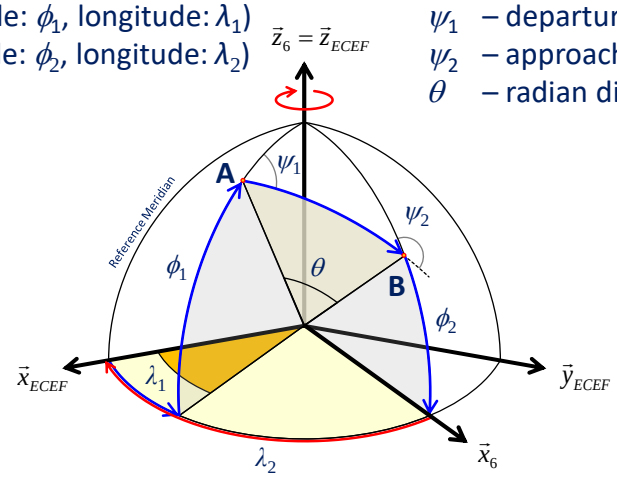

ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Rot$_y$(-θ)→Rot$_x$(ψ_2)→ Rot$_y$(ϕ_2)→Rot$_z$(-λ_2)→ECEF

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

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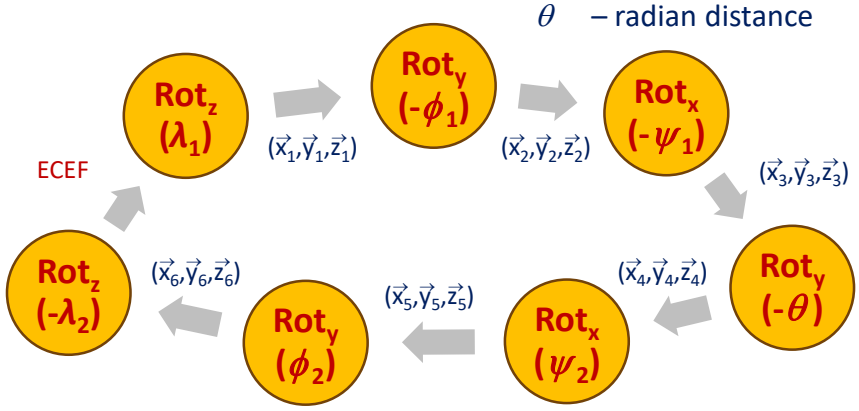

ECEF→Rot$_z$(λ_1)→Rot$_y$(-ϕ_1)→Rot$_x$(-ψ_1)→Rot$_y$(-θ)→Rot$_x$(ψ_2)→ Rot$_y$(ϕ_2)→Rot$_z$(-λ_2)→ECEF

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Finding the shortest path between two points on a sphere (orthodrome)

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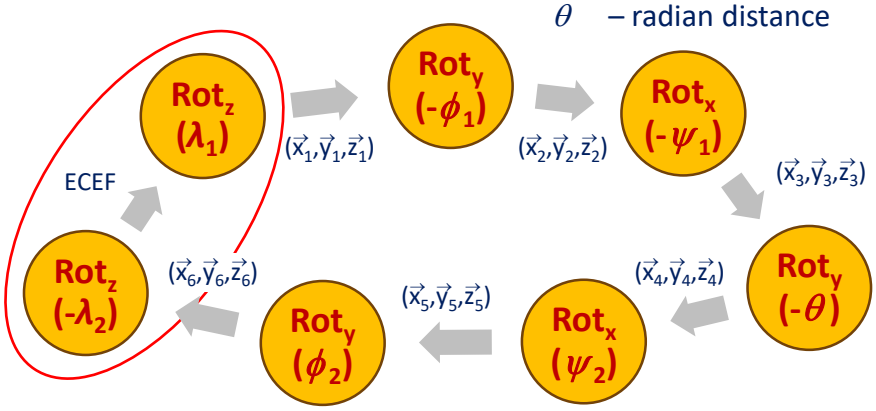

ECEF→Rot$_z$(λ_1)→Rot$_y$($-\phi_1$)→Rot$_x$($-\psi_1$)→Rot$_y$($-\theta$)→Rot$_x$(ψ_2)→ Rot$_y$(ϕ_2)→Rot$_z$($-\lambda_2$)→ECEF

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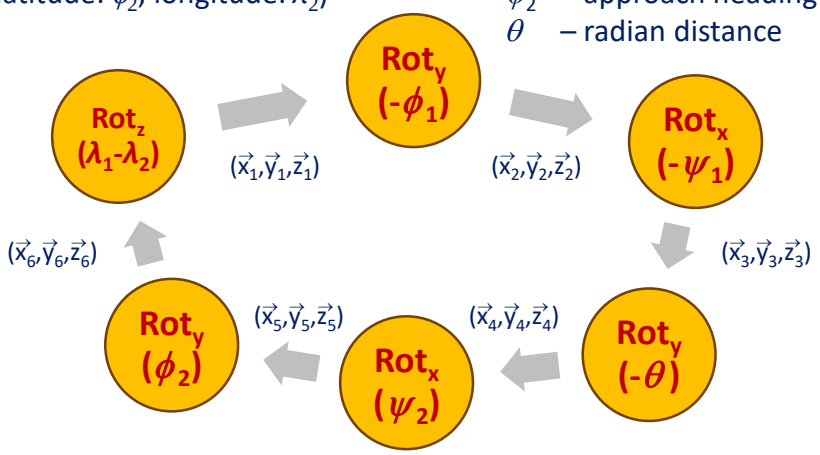

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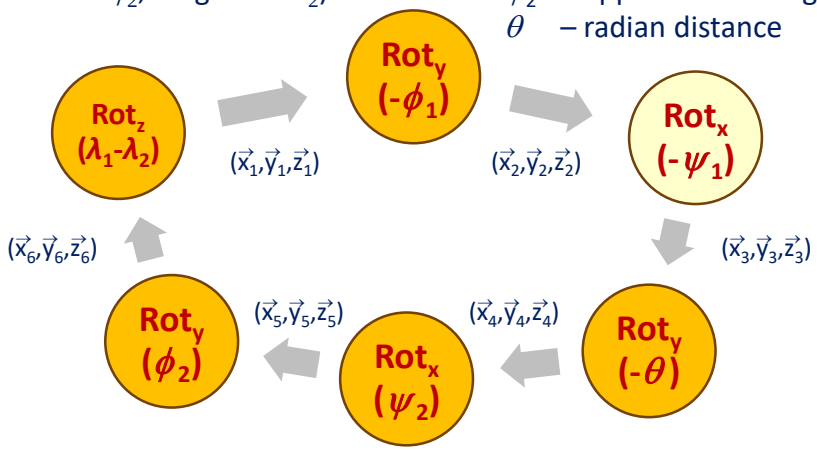

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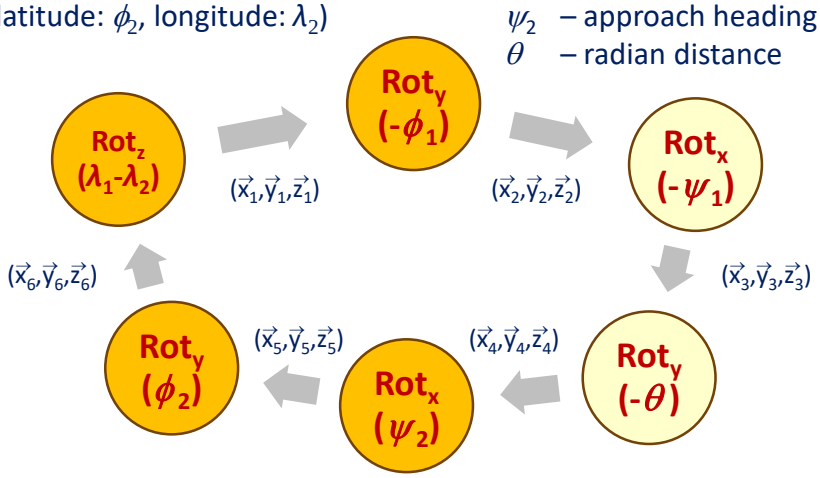

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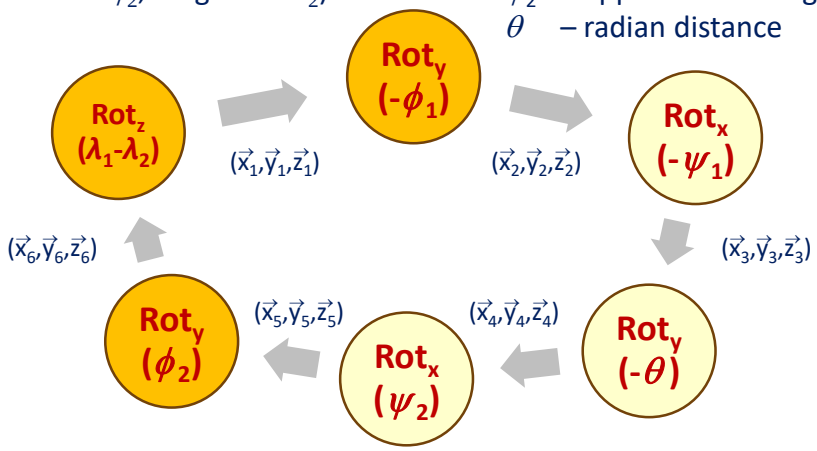

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A (latitude: ϕ_1, longitude: λ_1)
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$$\underbrace{\boxed{\mathrm{Rot}_x(-\psi_1)\cdot\mathrm{Rot}_y(-\theta)\cdot\mathrm{Rot}_x(\psi_2)}}_{M(\psi_1,\psi_2,\theta)}\cdot\underbrace{\boxed{\mathrm{Rot}_y(\phi_2)\cdot\mathrm{Rot}_z(\lambda_1-\lambda_2)\cdot\mathrm{Rot}_y(-\phi_1)}}_{N(\phi_1,\phi_2,\lambda_1-\lambda_2)}=I$$

$$M(\psi_1,\psi_2,\theta)\cdot N(\phi_1,\phi_2,\lambda_1-\lambda_2) = I \Leftrightarrow$$

$$\Leftrightarrow M(\psi_1,\psi_2,\theta) = N^{-1}(\phi_1,\phi_2,\lambda_1-\lambda_2) \Leftrightarrow \boxed{M(\psi_1,\psi_2,\theta) = N^{T}(\phi_1,\phi_2,\lambda_1-\lambda_2)}$$

N is a rotation matrix

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

$$M(\psi_1,\psi_2,\theta) = \mathrm{Rot}_x(-\psi_1)\cdot\mathrm{Rot}_y(-\theta)\cdot\mathrm{Rot}_x(\psi_2) =$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi_1 & \sin\psi_1 \\ 0 & -\sin\psi_1 & \cos\psi_1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi_2 & -\sin\psi_2 \\ 0 & \sin\psi_2 & \cos\psi_2 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ \sin\psi_1\sin\theta & \cos\psi_1 & \sin\psi_1\cos\theta \\ \cos\psi_1\sin\theta & -\sin\psi_1 & \cos\psi_1\cos\theta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi_2 & -\sin\psi_2 \\ 0 & \sin\psi_2 & \cos\psi_2 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta\sin\psi_2 & -\sin\theta\cos\psi_2 \\ \sin\psi_1\sin\theta & \cos\psi_1\cos\psi_2 + \sin\psi_1\cos\theta\sin\psi_2 & -\cos\psi_1\sin\psi_2 + \sin\psi_1\cos\theta\cos\psi_2 \\ \cos\psi_1\sin\theta & -\sin\psi\cos\psi_2 + \cos\psi_1\cos\theta\sin\psi_2 & \sin\psi_1\sin\psi_2 + \cos\psi_1\cos\theta\cos\psi_2 \end{bmatrix}$$

$$N(\phi_1,\phi_2,\lambda_1-\lambda_2) = \mathrm{Rot}_y(\phi_2)\cdot\mathrm{Rot}_z(\lambda_1-\lambda_2)\cdot\mathrm{Rot}_y(-\phi_1) =$$

$$= \begin{bmatrix} \cos\phi_2 & 0 & \sin\phi_2 \\ 0 & 1 & 0 \\ -\sin\phi_2 & 0 & \cos\phi_2 \end{bmatrix} \cdot \begin{bmatrix} \cos(\lambda_1-\lambda_2) & -\sin(\lambda_1-\lambda_2) & 0 \\ \sin(\lambda_1-\lambda_2) & \cos(\lambda_1-\lambda_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\phi_1 & 0 & -\sin\phi_1 \\ 0 & 1 & 0 \\ \sin\phi_1 & 0 & \cos\phi_1 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\phi_2\cos(\lambda_1-\lambda_2) & -\cos\phi_2\sin(\lambda_1-\lambda_2) & \sin\phi_2 \\ \sin(\lambda_1-\lambda_2) & \cos(\lambda_1-\lambda_2) & 0 \\ -\sin\phi_2\cos(\lambda_1-\lambda_2) & \sin\phi_2\sin(\lambda_1-\lambda_2) & \cos\phi_2 \end{bmatrix} \cdot \begin{bmatrix} \cos\phi_1 & 0 & -\sin\phi_1 \\ 0 & 1 & 0 \\ \sin\phi_1 & 0 & \cos\phi_1 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\phi_2\cos(\lambda_1-\lambda_2)\cos\phi_1 + \sin\phi_2\sin\phi_1 & -\cos\phi_2\sin(\lambda_1-\lambda_2) & -\cos\phi_2\cos(\lambda_1-\lambda_2)\sin\phi_1 + \sin\phi_2\cos\phi_1 \\ \sin(\lambda_1-\lambda_2)\cos\phi_1 & \cos(\lambda_1-\lambda_2) & -\sin(\lambda_1-\lambda_2)\sin\phi_1 \\ -\sin\phi_2\cos(\lambda_1-\lambda_2)\cos\phi_1 + \cos\phi_2\sin\phi_1 & \sin\phi_2\sin(\lambda_1-\lambda_2) & \sin\phi_2\cos(\lambda_1-\lambda_2)\sin\phi_1 + \cos\phi_2\cos\phi_1 \end{bmatrix}$$

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
ψ_2 – approach heading
θ – radian distance

$$\boxed{M(\psi_1,\psi_2,\theta) = N^T(\phi_1,\phi_2,\lambda_1-\lambda_2)}$$

$$M(\psi_1,\psi_2,\theta) = \mathrm{Rot}_x(-\psi_1)\cdot\mathrm{Rot}_y(-\theta)\cdot\mathrm{Rot}_x(\psi_2) =$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta\sin\psi_2 & -\sin\theta\cos\psi_2 \\ \sin\psi_1\sin\theta & \cos\psi_1\cos\psi_2 + \sin\psi_1\cos\theta\sin\psi_2 & -\cos\psi_1\sin\psi_2 + \sin\psi_1\cos\theta\cos\psi_2 \\ \cos\psi_1\sin\theta & -\sin\psi\cos\psi_2 + \cos\psi_1\cos\theta\sin\psi_2 & \sin\psi_1\sin\psi_2 + \cos\psi_1\cos\theta\cos\psi_2 \end{bmatrix}$$

$$N^T(\phi_1,\phi_2,\lambda_1-\lambda_2) = \left(\mathrm{Rot}_y(\phi_2)\cdot\mathrm{Rot}_z(\lambda_1-\lambda_2)\cdot\mathrm{Rot}_y(-\phi_1)\right)^T =$$

$$= \begin{bmatrix} \cos\phi_2\cos(\lambda_1-\lambda_2)\cos\phi_1 + \sin\phi_2\sin\phi_1 & \sin(\lambda_1-\lambda_2)\cos\phi_1 & -\sin\phi_2\cos(\lambda_1-\lambda_2)\cos\phi_1 + \cos\phi_2\sin\phi_1 \\ -\cos\phi_2\sin(\lambda_1-\lambda_2) & \cos(\lambda_1-\lambda_2) & \sin\phi_2\sin(\lambda_1-\lambda_2) \\ -\cos\phi_2\cos(\lambda_1-\lambda_2)\sin\phi_1 + \sin\phi_2\cos\phi_1 & -\sin(\lambda_1-\lambda_2)\sin\phi_1 & \sin\phi_2\cos(\lambda_1-\lambda_2)\sin\phi_1 + \cos\phi_2\cos\phi_1 \end{bmatrix}$$

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

A (latitude: ϕ_1, longitude: λ_1)
B (latitude: ϕ_2, longitude: λ_2)

ψ_1 – departure heading
ψ_2 – approach heading
θ – radian distance

$$M(\psi_1, \psi_2, \theta) = N^T(\phi_1, \phi_2, \lambda_1 - \lambda_2)$$

$\phi_1, \phi_2 \in [-\pi/2, \pi/2]$ $\quad \psi_1, \psi_2 \in [-\pi, \pi]$
$\lambda_1, \lambda_2 \in [-\pi, \pi]$ $\quad \theta \in]0, 2\pi[$

$M_{11} = \cos\theta$ $\qquad (N^T)_{11} = \cos\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 + \sin\phi_2 \sin\phi_1$

$$\cos\theta = \cos\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 + \sin\phi_2 \sin\phi_1$$

$M_{21} = \sin\psi_1 \sin\theta$ $\qquad (N^T)_{21} = -\cos\phi_2 \sin(\lambda_1 - \lambda_2)$

$M_{31} = \cos\psi_1 \sin\theta$ $\qquad (N^T)_{31} = -\cos\phi_2 \cos(\lambda_1 - \lambda_2)\sin\phi_1 + \sin\phi_2 \cos\phi_1$

$(\sin\theta \neq 0)$

$$\tan\psi_1 = \frac{-\cos\phi_2 \sin(\lambda_1 - \lambda_2)}{-\cos\phi_2 \cos(\lambda_1 - \lambda_2)\sin\phi_1 + \sin\phi_2 \cos\phi_1}$$

$M_{12} = -\sin\theta \sin\psi_2$ $\qquad (N^T)_{12} = \sin(\lambda_1 - \lambda_2)\cos\phi_1$

$M_{13} = -\sin\theta \cos\psi_2$ $\qquad (N^T)_{13} = -\sin\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 + \cos\phi_2 \sin\phi_1$

$(\sin\theta \neq 0)$

$$\tan\psi_2 = \frac{-\sin(\lambda_1 - \lambda_2)\cos\phi_1}{\sin\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 - \cos\phi_2 \sin\phi_1}$$

MATLAB hint: use `atan2` to compute ψ_1 and ψ_2.

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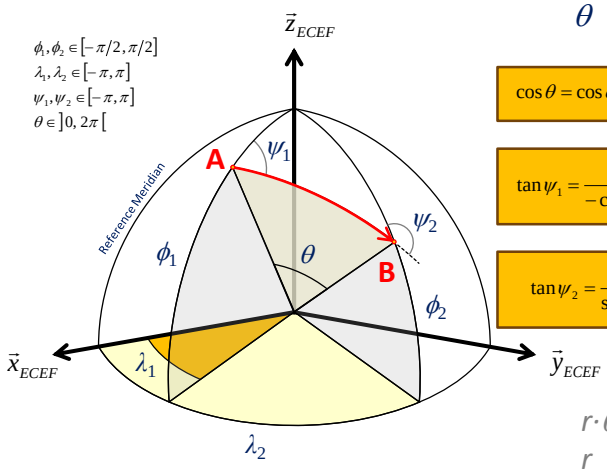
$\lambda_1, \lambda_2 \in [-\pi, \pi]$
$\psi_1, \psi_2 \in [-\pi, \pi]$
$\theta \in]0, 2\pi[$



$$\cos\theta = \cos\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 + \sin\phi_2 \sin\phi_1$$

$$\tan\psi_1 = \frac{-\cos\phi_2 \sin(\lambda_1 - \lambda_2)}{-\cos\phi_2 \cos(\lambda_1 - \lambda_2)\sin\phi_1 + \sin\phi_2 \cos\phi_1}$$

$$\tan\psi_2 = \frac{-\sin(\lambda_1 - \lambda_2)\cos\phi_1}{\sin\phi_2 \cos(\lambda_1 - \lambda_2)\cos\phi_1 - \cos\phi_2 \sin\phi_1}$$

MATLAB hint: use `atan2` to compute ψ_1 and ψ_2.

$r \cdot \theta$ – orthodrome length
r – Earth radius

Great Circle Navigation

Finding the shortest path between two points on a sphere (orthodrome)

- Orthodromes are the shortest path, on the surface of sphere, between two points;
- Orthodromes follow great circles;
- Great circles share the center with the sphere;
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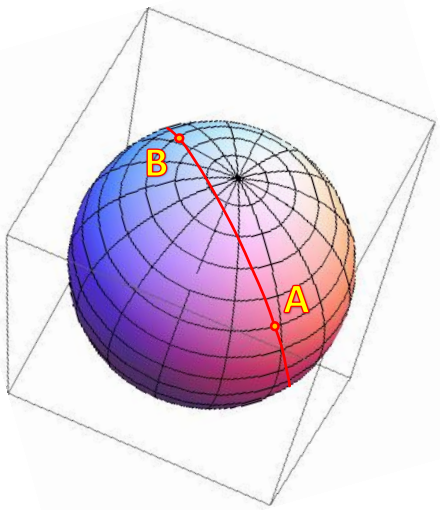
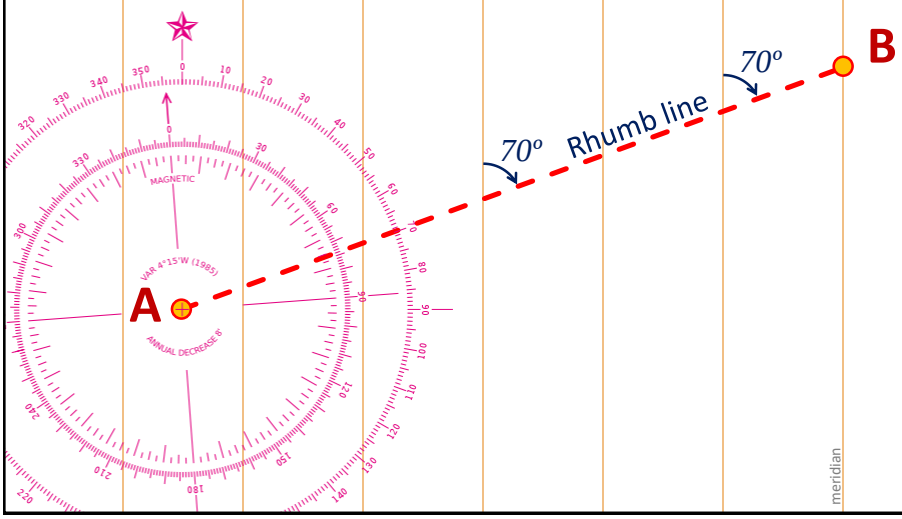
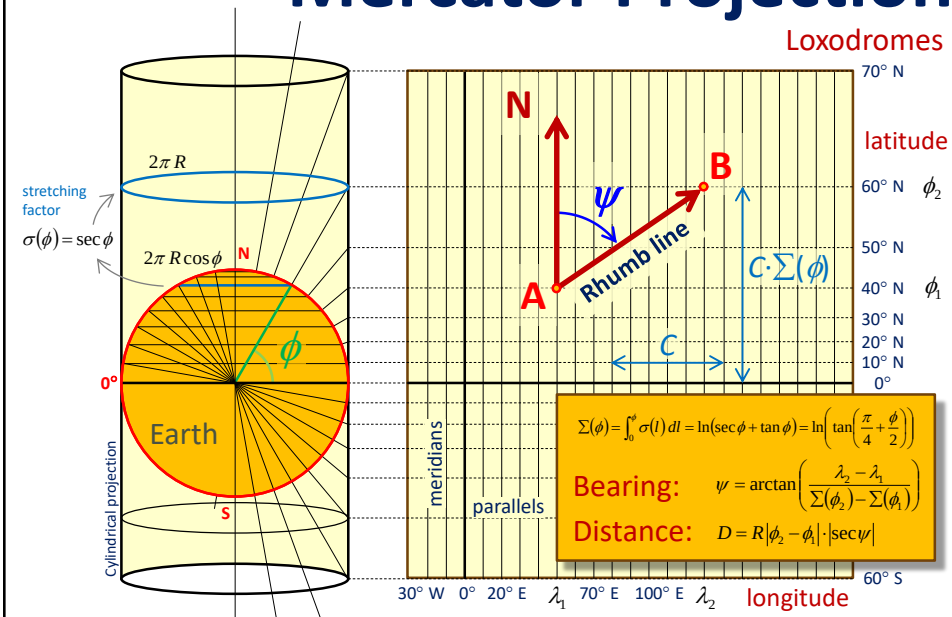


Keeping on Course

Following a rhumb line (loxodrome)



Mercator Projection



$$\Sigma(\phi) = \int_0^{\phi} \sigma(l)\,dl = \ln(\sec\phi + \tan\phi) = \ln\left(\tan\left(\frac{\pi}{4} + \frac{\phi}{2}\right)\right)$$

Bearing: $\psi = \arctan\left(\dfrac{\lambda_2 - \lambda_1}{\Sigma(\phi_2) - \Sigma(\phi_1)}\right)$

Distance: $D = R\left|\phi_2 - \phi_1\right| \cdot \left|\sec\psi\right|$

James Alexander, "Loxodromes: A Rhumb Way to Go," Mathematics Magazine, Vol. 77, No. 5, December 2004, pp.349-356

MATLAB hint: use `atan2` to compute ψ.